Optimization of Motor

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Note: Kindly refer here (https://ajaygunalan.github.io/blog/notes/motor/motor.pdf) for the various definition and derivation. Here the focus is only to understand which factors need to be maximised and minimised.

Introduction

What is a robot?

A robot is a bunch of actuator connected by a link(a structure). Actuators, being one of the fundamental element of a robot, will determine the core capabilities and limitation of a robot. Thus a thorough understanding of the actuator will help us to optimise it based upon our requirment. In this article we restrict ourself to Electromagnetic Actuator (Motors), since it is the most widely used one and has a higher energy efficiency compared to others like hydraulic, pneumatic, etc. Out of all the various type of motor like Induction, Reluctance, PMSM/BLDC, DC, Stepper, etc. Our goal is to understand various factors of EM actuator which influences the perfomance and hopefully to justify why **pancake shaped, Quasi Direct-Drive, Permanent Magnet Synchronous Motors(PMSM), with 4Q control drivers and elastic element** will be the norm of the robotics in future.

*Note: (PMSM) is physically same as the BLDC Motor. We call it PMSM/BLAC when we apply a sinuosoidal emf and BLDC when we apply trapezoidal emf.

Why PMSM ? Why not DC, Coreless DC ?

Well, brushless means reduced noise, wear, and friction caused by brushes and torque-speed characteristics is linear[1] and high energy efficiency make it a attractive choice.

Pan-Cake Shape ?

Motor Parameters

Parameterisation means, to develop the model of a "system" interms of paramters(variable) and to see how various parameters affect the "system". The

developed model should be verified by comparing the theoritical prediction form the model against experimental data and based upon the error, we should improve our model till we are satisfied with model's accuracy.

For a 3-phase BLDC/PMSM, the appropriate model is:

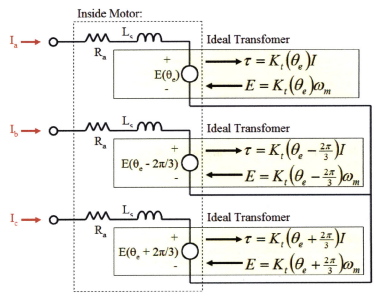

Figure 1: Model of a 3-phase BLDC/PMSM, Source:[2]

Selection Criteria

[3]

A smaller motor will have lower iron losses at a given speed (less iron to create drag), and a larger motor will have lower copper losses for a given torque.

One interesting approach to motor sizing for a robot is to choose a small motor sized for walking, and then to overload it heavily during more strenuous activities.

This might result in huge copper losses due to squaring the high winding current, but for brief periods the effect on overall efficiency is minimal.

The thermal mass of the motor windings is high enough that it might not burn up for several seconds, and if it is cooled well enough the power level may be sustainable for longer activities.

This is particularly true for permanent magnet brushless motors, in which the stator coils are stationary and in contact with the case, well-positioned for heat sinks or water cooling jackets. (For brushed motors the magnets are on the outside and the windings are in the rotor, leaving few options for cooling.

Perfomance

To Do

1. Finish 1.3.3 and 1.4.2 in [4]
2. Maxon Document
3. Asuting BLDC chapter
4. BLDC and chaos

References

[1] H. Asada and K. Youcef-Toumi, *Direct-drive robots: Theory and practice*. Cambridge, Mass: MIT Press, 1987.

[2] S. W. (. W. Colton, "Design and prototyping methods for brushless motors and motor control," Thesis, Massachusetts Institute of Technology, 2010 [Online]. Available: http://dspace.mit.edu/handle/1721.1/61599

[3] S. Seok, A. Wang, D. Otten, and S. Kim, "Actuator design for high force proprioceptive control in fast legged locomotion," in *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2012, pp. 1970–1975 [Online]. Available: http://ieeexplore.ieee.org/document/6386252/. [Accessed: 12-Sep-2018]

[4] J. R. Hendershot and T. J. E. Miller, *Design of brushless permanent-magnet machines*. Venice, Florida: Motor Design Books, 2010.